

Perfect Mirror Effects in Metasurfaces of Silicon Nanodisks at Telecom Wavelength

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This article explores the design and optimization of nanodisk metasurfaces for achieving high reflectivity at a defined wavelength. The telecom wavelength of 1550 nm is particularly focused, selected for its potential applications in next-generation gravitational wave detectors. At this wavelength, the research goes toward the development of thin, low-loss, high-reflective coatings, where the metasurface can be chosen as an alternative. An optimization process for the dimensional parameters of nanodisks is proposed based on a systematic tuning approach, which facilitates the realization of various configurations of high-reflective metasurfaces. The concept of the “magnetic mirror effect” is examined in detail, where the magnetic dipole resonance aligns with the anapole state. Additionally, high reflectivity at the electric dipole resonance (“electric mirror effect”) and at the excitation of several multipole moments is explored, including high-order modes. This variety of configurations affords more flexibility in the phase manipulation of the reflected beam. Furthermore, the potential experimental realization of mirror effects is discussed by exploring the structure in the surrounding medium with a refractive index of $n_d = 1.4$. This research platform provides a promising tool for the fabrication of high-reflective nanodisk metasurfaces and demonstrates its applicability across various fields.

requirements. This involves a diversified and in-depth exploration of nanoparticle configurations and their operational modes.^[2,3]

The motivation for our metasurface design has its origin in the field of gravitational wave astronomy. Our research focus extends towards next-generation projects, such as the Einstein Telescope,^[4,5] where new techniques and materials are expected to be used at the telecom wavelength (1550 nm) to increase the low-frequency sensitivity of the gravitational wave detectors. Both table-top and large-scale experiments must employ low-loss highly reflective coatings deposited on crystalline structures.^[6]

Previous research demonstrated possibilities to design reflective metasurfaces made of periodically arranged silicon nanospheres on top of a sapphire substrate and provided consequent analysis of reflection sensitivity to fabrication limitations.^[7,8] The design of the nanospherical metasurface has the advantage that the behavior of the system is determined by a minimum

set of dimensional parameters. However, nanosphere manufacturing technology, such as laser printing,^[9] faces challenges in maintaining uniform size and precise positioning for each nanoparticle. As an alternative, electron beam lithography (EBL) is considered. The EBL capabilities are constrained by the fabrication of vertical walls and, thus, are well suited for the fabrication of cylinders, cuboids, triangular prisms, etc. To circumvent

1. Introduction

The appearance of thin, lightweight, optically efficient devices to control and manipulate light has recently been widely acknowledged across diverse scientific disciplines.^[1] Considerable research efforts are directed toward the development of metasurfaces capable of meeting stringent optical and mechanical

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the limitations associated with the uncertainties in achieving the perfect shape of the sharp corners,^[10] we choose for our investigation a cylindrical nanoparticle configuration, which is explored in this paper.

The dominant resonant effect occurring in nanocylinders is Mie resonance, based on the same mechanisms as in nanospheres. High reflectivity from nanocylinders can be observed at either electric or magnetic dipole resonances.^[11] However, the relative spectral positioning of the dipole resonances for a sphere is strictly fixed, whereas the ratio of a cylinder height to its radius can alter their mutual spectral correlation.^[12,13] For thin nanocylinders resembling a nanodisk shape (when their height, H , is less than the diameter, D), the magnetic dipole resonance is spectrally located in close proximity to the electric dipole resonance, whereas for tall nanocylinders ($H > D$) the resonances are spectrally spaced far apart.

The preference toward structures consisting of nanodisks ($H < D$) is motivated by achieving thinner and lighter coatings for applications in quantum metrology and gravitational wave physics. At the same time, this choice helps to mitigate the challenges associated with numerous manufacturing defects.^[14,15] However, the optical properties of metasurfaces made from nanodisks require a more detailed investigation.

At certain size parameters of the nanodisks, the electric and magnetic dipole resonances, excited by external waves, spectrally overlap. If this occurs, it may lead to destructive interference between the electric and magnetic dipole backscattering (Kerker effect), and consequently to suppression of reflection.^[16,17] To avoid this behavior and obtain high reflectivity, one should choose a geometry where the dipole resonances are spectrally separated.

By a large reduction in the height of a nanodisk (height is less than radius), the magnetic dipole (MD) resonance is placed in the shorter wavelength range compared to the electric dipole resonance. Another interesting point for such disk configuration is the radiationless state of the electric dipole (ED) moment, called anapole state.^[18,19] The reflective properties of the metasurface in this state are completely defined by other multipole contributions. Positioning the MD resonance at this state allows to obtain ideal magnetic dipole scattering^[20] and design a “magnetic mirror” with zero phase shift of reflected electric field in relation to the incident light wave.^[21] This paper provides a detailed discussion of the construction method for a “magnetic mirror” tailored to the operational wavelength of 1550 nm.

Furthermore, the reflection of the mirror can also be predominantly characterized by the contribution of only the ED moment, essentially forming an “electric mirror”. We will show how specific dimensional parameters of the metasurface can be easily adjusted to operate the mirror at the ED moment or any other mode, with the potential to change the phase of the reflected beam. A comprehensive discussion of all cases, accompanied by information on potential future experimental realization, will be presented in the concluding sections.

2. Analytical Approach

High-refractive-index dielectric nanoparticles support electric and magnetic multipole resonances, providing amplitude and phase manipulations with the incident light field. The shape of nanocylinders, in particular the ratio between radii and height,

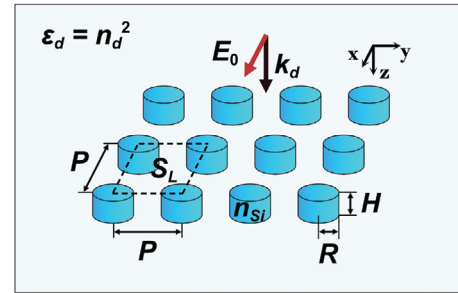


Figure 1. The infinite array of silicon nanocylinders, characterized by their height H , radius R and periodicity P in x - and y -directions. The unit cell area, defined by the square of periodicity, is denoted as $S_L = P^2$. The refractive index of nanocylinders is n_{Si} . The array is embedded in the surrounding medium with a dielectric constant $\epsilon_d = n_d^2$, where n_d is the refractive index of medium. The structure is perpendicularly exposed to a monochromatic plane wave with a linearly polarized electric field E_0 along the x -axis.

influences the location of electric and magnetic dipole (multipole) resonances on the spectrum, spacing the resonances apart from each other or making them overlap.^[12]

Let's examine a metasurface, represented as a 2D infinite periodic array composed of identical nanoparticles (**Figure 1**). The metasurface is normally irradiated by linearly polarized light, with the electric field E_0 oriented along the x axis. Within the coupled-dipole approximation theory, the electric-field reflection r and transmission t coefficients can be written as Ref. [22]

$$r = \frac{ik_d}{E_0 2S_L \epsilon_0 \epsilon_d} \left[p_x - \frac{1}{v_d} m_y \right] \quad (1)$$

$$t = 1 + \frac{ik_d}{E_0 2S_L \epsilon_0 \epsilon_d} \left[p_x + \frac{1}{v_d} m_y \right] \quad (2)$$

where k_d is the wave number in a surrounding medium, S_L is the unit cell area ($S_L = P^2$ for the square lattice, where P is the array periodicity), ϵ_0 is the vacuum permittivity, ϵ_d is the permittivity of a surrounding medium, v_d is the speed of light in the surrounding medium, p_x and m_y are the in-plane components of electric and magnetic dipole moments, respectively, induced in all particles by the incident wave. Here, the electric field E_0 is taken in the plane of the metasurface.

High reflectance is typically attained at the resonances of electric and magnetic dipole moments when they are spectrally separated from one another.^[11] For this demonstration, let's present the dipole moments in the metasurface using the tensors of electric $\hat{\alpha}^{\text{eff-E}}$ and magnetic $\hat{\alpha}^{\text{eff-M}}$ effective dipole polarizabilities,

$$p_x = \epsilon_0 \epsilon_d \alpha_{\parallel}^{\text{eff-E}} E_0 \quad m_y = \alpha_{\parallel}^{\text{eff-M}} H_0 \quad (3)$$

where $\alpha_{\parallel}^{\text{eff-E}}$ and $\alpha_{\parallel}^{\text{eff-M}}$ are the in-plane components of the corresponding polarizability tensors, E_0 and H_0 are electric and magnetic fields of the incident plane waves at the points of dipole locations, $H_0 = \sqrt{\epsilon_0 \epsilon_d / \mu_0} E_0$ (here μ_0 is the vacuum magnetic permeability). Note that the effective polarizabilities can differ significantly from single particle polarizabilities due to the interparticle coupling in arrays, especially under the condition of collective lattice resonances.^[23] However, if the single particle

dipole resonances are realized at subdiffractional periodicity, the effect of interparticle coupling on the polarizabilities is weak in this spectral range. This results in only small spectral shifts of resonances, while the collective lattice resonances are excluded.^[23,24]

Using (3), the reflection and transmission coefficients can be written as

$$r = \frac{ik_d}{2S_L} [\alpha_{\parallel}^{\text{eff}-E} - \alpha_{\parallel}^{\text{eff}-M}] \quad (4)$$

$$t = 1 + \frac{ik_d}{2S_L} [\alpha_{\parallel}^{\text{eff}-E} + \alpha_{\parallel}^{\text{eff}-M}] \quad (5)$$

Note that at the resonant wavelength λ_R , the real part of the dipole polarizability is zero, so that the polarizability is purely defined by the imaginary part.

Let's determine the conditions for achieving 100% reflectivity in a system with electric and magnetic dipole responses. From (5) and requirement $t = 0$ we obtain

$$\alpha_{\parallel}^{\text{eff}-E} = \frac{i2S_L}{k_d} - \alpha_{\parallel}^{\text{eff}-M} \quad (6)$$

and from (4) and requirement $|r| = 1$ (i.e., $r = e^{i\phi}$, $\phi \in [-\pi; \pi]$), the general solution is

$$\alpha_{\parallel}^{\text{eff}-M} = \frac{iS_L}{k_d} (1 + e^{i\phi}), \quad \alpha_{\parallel}^{\text{eff}-E} = \frac{iS_L}{k_d} (1 - e^{i\phi}) \quad (7)$$

It is important to note that control of the contributions of electric and magnetic dipole polarizabilities in total reflection, in principle, makes it possible to control the phase of reflection fields in the range of 0 to 2π .^[25] The phase angle ϕ can be considered as the phase difference between the reflected and incident electric fields.

Three partial cases can be distinguished:

$$1) \text{ if } r = 1, (\phi = 0) \text{ then } \alpha_{\parallel}^{\text{eff}-M} = i2S_L/k_d \text{ and } \alpha_{\parallel}^{\text{eff}-E} = 0$$

(magnetic mirror effect) (8)

$$2) \text{ if } r = -1, (\phi = \pi) \text{ then } \alpha_{\parallel}^{\text{eff}-M} = 0 \text{ and } \alpha_{\parallel}^{\text{eff}-E} = i2S_L/k_d$$

(electric mirror effect) (9)

$$3) \text{ if } r = \pm i, (\phi = \pm\pi/2) \text{ then } \alpha_{\parallel}^{\text{eff}-M} = (i \mp 1)S_L/k_d \text{ and}$$

$$\alpha_{\parallel}^{\text{eff}-E} = (i \pm 1)S_L/k_d \quad (10)$$

The first two cases correspond to a single-dipole response, when the polarizability of the other dipole is negligibly small and does not contribute to reflection and transmission. Case 1) is realized due to the radiation of a magnetic dipole, resulting in the reflection of an electric field without a phase jump (the effect of a magnetic mirror). Case 2) corresponds to the effect of an electric mirror with a π -change in the phase of the reflected electric field. For case 3), the participation of both dipoles is necessary: total reflection is realized between electric and magnetic dipole resonances under special conditions $\text{Im}(\alpha_{\parallel}^{\text{eff}-M}) = \text{Im}(\alpha_{\parallel}^{\text{eff}-E}) = S_L/k_d$

and $\text{Re}(\alpha_{\parallel}^{\text{eff}-M}) = -\text{Re}(\alpha_{\parallel}^{\text{eff}-E}) = \pm S_L/k_d$. It is important to note the absorption in the system is assumed to be zero.

Thus, from the above analysis we can write down two necessary and sufficient conditions to achieve total reflection in the ED-MD approximation for the metasurfaces under consideration:

$$\text{Re}(\alpha_{\parallel}^{\text{eff}-M}) = -\text{Re}(\alpha_{\parallel}^{\text{eff}-E}), \quad \text{Im}(\alpha_{\parallel}^{\text{eff}-M}) + \text{Im}(\alpha_{\parallel}^{\text{eff}-E}) = \frac{2S_L}{k_d} \quad (11)$$

Under these conditions, the phase ϕ of the reflected electric field is determined by the expression:

$$\tan \phi = \frac{\text{Re}(\alpha_{\parallel}^{\text{eff}-E}) - \text{Re}(\alpha_{\parallel}^{\text{eff}-M})}{\text{Im}(\alpha_{\parallel}^{\text{eff}-M}) - \text{Im}(\alpha_{\parallel}^{\text{eff}-E})} \quad (12)$$

3. Spectral Tuning of the Magnetic Mirror with the Electric-Dipole Anapole State

Let us consider case 1), when the total reflection is achieved due to the magnetic dipole resonance excited against the background of suppression of electric dipole polarizability in the anapole state, that is, at the point where $\alpha_{\parallel}^{\text{eff}-E} = 0$. This approach is justified by the fact that a significant, close to zero suppression of the electric dipole moment can only be achieved in the anapole state, which is basically the result of destructive interference between the quasi-static electric dipole and the toroidal dipole moments.^[18,26] As a first step, we will show how the magnetic dipole resonance of a single silicon disk can be tuned to its electrical dipole-anapole state in the telecom spectral range for $\lambda = 1550$ nm. All numerical steps are conducted using the commercial software Comsol Multiphysics, which is based on the finite element method. The refractive index of silicon nanoparticles in Section 3 and Section 4.1 was set to be wavelength independent and equal to its value for crystalline silicon at the design wavelength: $n_{\text{Si}}(\lambda) = n_{\text{Si}}(1550) = 3.48$.^[27] After this, having chosen a periodicity, we will construct a metasurface from these disks and demonstrate the effect of an ideal magnetic mirror at $\lambda = 1550$ nm precisely due to the coincidence of the magnetic dipole resonance with the anapole state.

Here, we consider an environment with a refractive index $n_d = 1$. In the final part of the paper, the results obtained will be generalized to the case of an environment with a higher refractive index.

3.1. Tuning of the Magnetic Resonance to the Anapole State for a Single Disk Particle

Single spherical nanoparticles support spectrally separated resonant optical responses, with the magnetic resonance occurring at a longer wavelength range compared to the electric resonance.^[23] A similar behavior is observed for a nanocylinder with a height equal to twice the radius ($H = 2R$). To illustrate this statement, let's consider both cylindrical and spherical nanoparticles with the same radius $R = R_{\text{sp}}$, where R and R_{sp} are the radius of a cylinder and a sphere, respectively. As a starting value, R is estimated from the MD resonant condition for a sphere in Mie theory $R \approx$

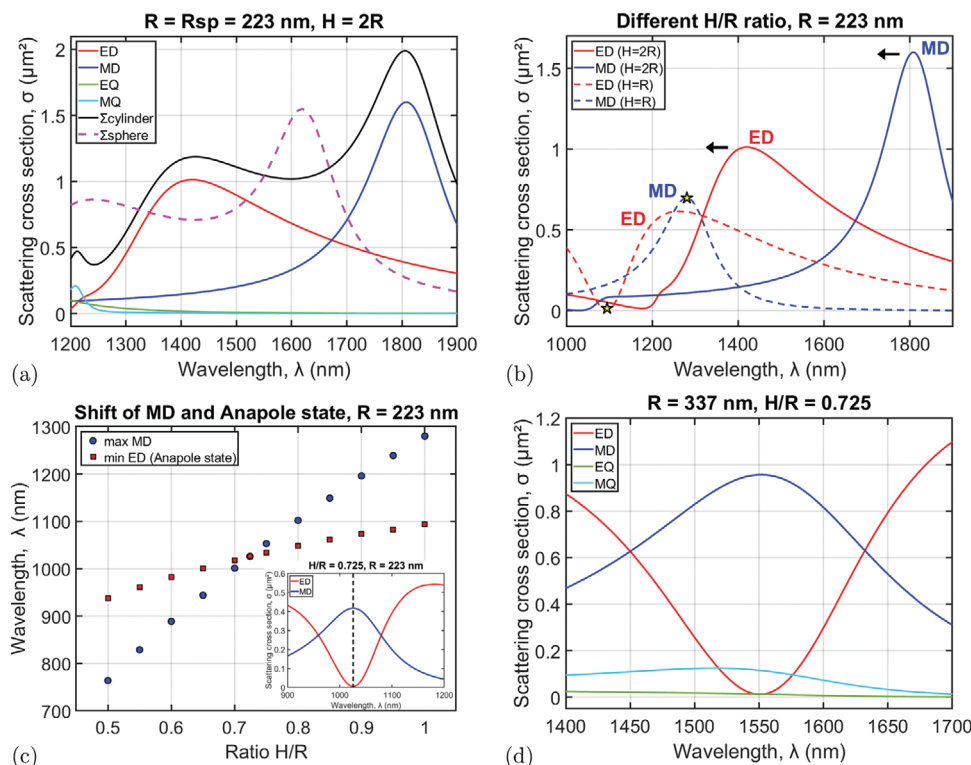


Figure 2. a) Scattering cross-section (SCS) and multipole decomposition calculated for a nanocylinder with radius $R = 223$ nm and height $H = 2R$. Shown is the comparison with the SCS of a nanosphere of the same radius ($R = R_{\text{sp}}$). b) Redistribution of the magnetic and electric dipole resonances in the spectra of SCSs by changing the height of the nanodisk at a fixed radius $R = 223$ nm (changing the aspect ratio). The dipole resonances overlap when $H \approx R$. The asterisk symbols denote the maximum of magnetic dipole contribution and the minimum of the electric dipole response (the anapole state). c) The spectral position of the MD maximum and anapole state as a function of the aspect ratio of the nanodisk. These states intersect at $H/R = 0.725$. The multipole decomposition of SCS for this point is plotted in the lower right corner of the graph. d) The alignment of the MD resonance and the anapole state to a 1550 nm wavelength by tuning the radius to $R = 337$ nm at the fixed aspect ratio of $R/H = 0.725$. All calculation is done with a constant refractive index of silicon $n_{\text{Si}} = 3.48$.

$\lambda/2n_{\text{Si}}$, where $\lambda = 1550$ nm, $n_{\text{Si}} = 3.48$, resulting in a radius of approximately 223 nm.

The total scattering cross-section (SCS), along with its multipole decomposition for a nanocylinder, are presented in **Figure 2a** by solid curves (see Appendix for the details of the calculation). The contribution of quadrupole moments (EQ, MQ) to the total scattering is found to be negligible in comparison to the electric (ED) and magnetic (MD) dipole moments. As expected, the total SCS of the nanosphere (dashed curve in **Figure 2a**) has similar shape as that of the nanocylinder, but blueshifted due to the smaller volume of the nanosphere.

In conclusion, for nanocylinders whose height surpasses their diameter, we operate exclusively in the distinct dipole regime described by the MD resonance at longer wavelengths.

Before further discussion, let's pre-define the terminology for nanocylinders with a height less than diameter ($H/R < 2$), calling them nanodisks.

By reducing the nanodisk height while keeping the radius constant, or in other words, by altering the height-to-radius aspect ratio (**Figure 2b**), both the ED and MD resonances shift toward shorter wavelengths. Notably, the MD peak shifts faster compared to the ED peak. The tendency can be explained by electric field redistribution within the particle, where MD has a peripheral distribution of polarizability currents.^[13] In contrast, the ED

has a concentric field inside of the nanoparticle, making it comparatively less sensitive to changes in height.

As the nanodisk height decreases, the resonances come into closer proximity. In the regime around equal height and radius ($H \approx R$), the dipole resonances overlap, leading to the resonant suppression of the backward scattering known as the Kerker effect.^[11,13] Continuing the reduction in height results in the resonances drifting once more apart, with the MD peak positioned to the shorter wavelength relative to the ED one. The maximum of the MD resonance is marked by the asterisk symbol in **Figure 2b**. In this regime, particular attention is drawn to the anapole state, also denoted by the asterisk symbol placed on the left side from the ED peak. The anapole state features substantial suppression of the ED contribution, thereby supporting the reflectivity solely due to the MD response.^[20]

Figure 2c illustrates the shift of the anapole state and the maximum of the MD contribution with respect to the height-to-radius aspect ratio (H/R). Their intersection defines the exact ratio when the resonant maximum of the magnetic dipole moment overlaps the anapole state. Such a point corresponds to the ratio $H/R \approx 0.725$ at the wavelength of $\lambda = 1026$ nm. In order to achieve the same condition at the wavelength of interest (1550 nm), the size of nanodisk should be scaled accordingly. The multipole decomposition of SCS for the scaled disk of the radius $R = 337$ nm and

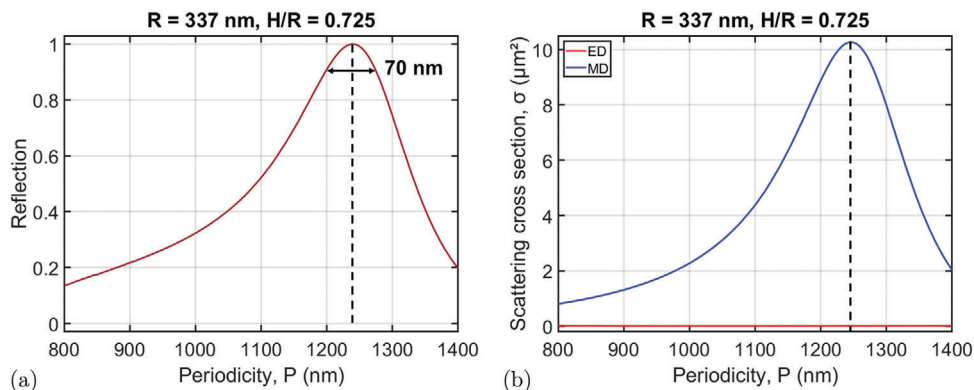


Figure 3. The behavior of the array of identical nanodisks ($R = 337$ nm, $H/R = 0.725$) at 1550 nm wavelength. a) The dependence of reflection on periodicity. Full reflection occurs around $P = 1239.5$ nm. The tolerance in the periodicity can be up to 70 nm for the reflection of more than 90%. b) Dipole decomposition in the periodicity range. There is no mutual contribution to the anapole state of each individual nanoparticle. The magnetic dipole has its maximum at the same periodicity $P = 1239.5$ nm as total reflection.

the aspect ratio $H/R = 0.725$ verifies the pure MD resonance without ED contribution at the wavelength of 1550 nm (Figure 2d). Although the MQ moment is also shifted into the telecom spectral range, its scattering cross-section is significantly smaller than for the MD moment.

3.2. Magnetic Mirror Effect

The array of nanodisks forms a metasurface characterized by its reflectivity. Magnetic mirror metasurface features high reflectance at the magnetic dipole resonance without any electric dipole excitation. Moreover, at such an operational point, the incident light beam reflects back without any phase shift in the absence of contribution from higher order modes, as seen in Equation (8).

Transitioning from the analysis of a single nanoparticle to the structured collective behavior, it is essential to account for the interparticle coupling effect. The position of nanodisks, described by periodicity P (Figure 1), has an impact on the total electric field of nanoparticles. This influence results from the mutual interactions in combination with the individual response to the incident light field.

A single nanoparticle resonating at the MD moment aligned to the anapole state at a wavelength of 1550 nm has geometric parameters of radius $R = 337$ nm and aspect ratio $R/H = 0.725$. Taking these parameters for each nanodisk of the array, we observe the change in reflection as a function of periodicity (Figure 3a). The full reflectance in the case of zero absorption for the non-diffractive periodicity range is obtained at $P = 1239.5$ nm.

By extracting the polarization density for a nanoparticle within the array and incorporating it into the equations for the multipole moments, and then calculating the scattering cross-section using Equation (A2) for a single nanoparticle in a homogeneous surrounding, we formally obtain information about which multipoles make a resonant contribution to the perfect reflection effect. This type of analysis has been applied to the current metasurface (Figure 3b), showing the dependency of scattering cross-section on periodicity. The plot demonstrates that the MD exhibits the same profile as the reflection curve, with its maximum resonance situated at the identical periodicity of $P = 1239.5$ nm.

The electric dipole moment, on the other hand, is suppressed and remains unaffected by periodicity change.

The resonance quality factor, defined as the periodicity misalignment tolerable to exceed 90% reflection, is 70 nm.

In this section, we showed how to conduct an optimization procedure in order to find the parameters for the magnetic mirror effect at a specific wavelength. In the next section, we consider the electric mirror effect and other possible solutions for nanodisk metasurfaces.

4. Ideal Reflection of Nanodisk Metasurface with Various Phase Control

4.1. Electric Mirror Effect

The realization of a fully reflective mirror is possible not only at the magnetic dipole resonance but also at the electric one Equation (9). It's important to note that the ideal electric mirror effect is achievable in the complete absence of a magnetic response. Although such nanoparticle geometries do not allow for a radiationless state of the magnetic dipole moment, it is still feasible to tune the metasurface in such a way that the input from the MD is negligible compared to the ED moment.

We know that by reducing the radius of the nanodisk and keeping the aspect ratio the same, we reduce the volume of the nanoparticle and shift electric and magnetic resonances simultaneously to a smaller wavelength range. To get the electric mirror effect at 1550 nm wavelength, the radius of the nanodisk is varied by keeping the aspect ratio of nanoparticles and periodicity of the array the same (Figure 4a). The full reflection for the electric mirror effect is observed at $R = 308$ nm. To show that the following analysis is correct, the multipole decomposition is presented in Figure 4b. The resonance of ED on the plot is placed at 1550 nm wavelength, with a subtle maximum shift attributed to the influence of interparticle coupling.

The phenomenon known as the electric mirror effect is characterized by a phase shift of the reflected beam with respect to the incident light field to $\phi = \pi$, whereas the phase shift for the magnetic mirror effect is equal to zero, $\phi = 0$. The standing-wave pattern in Figure 4c shows that the π -phase shift for the

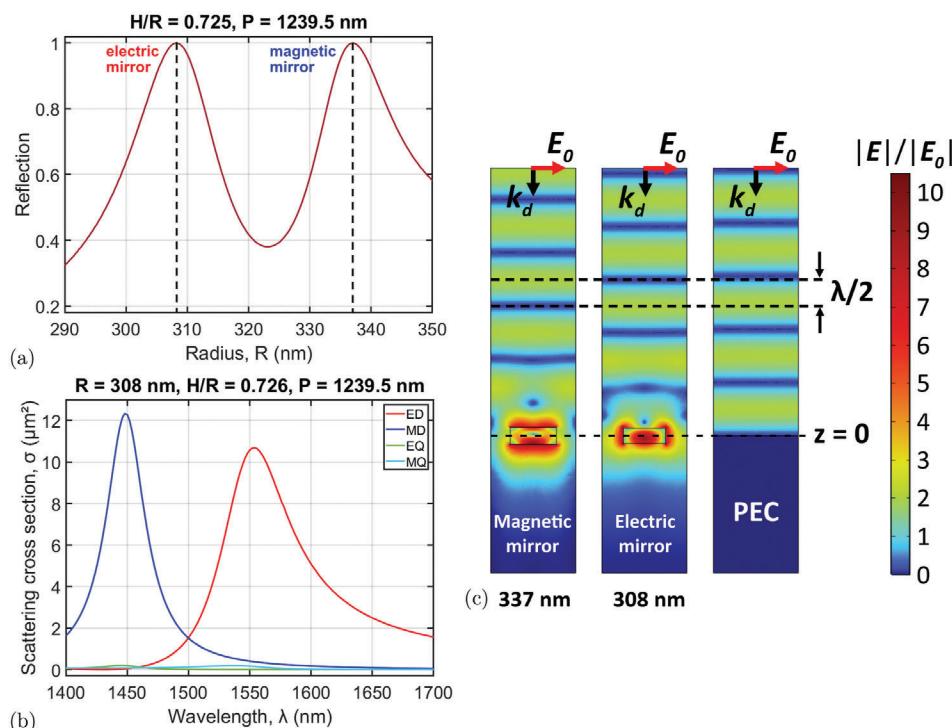


Figure 4. a) Reflection of different metasurfaces at 1550 nm wavelength versus disk radius (at fixed height-to-radius aspect ratio). The decrease of the radius leads to the transition from the mirror with magnetic effect ($R = 337$ nm, $H/R = 0.725$, $P = 1239.5$ nm) to the mirror operating at electric dipole resonance ($R = 308$ nm, $H/R = 0.725$, $P = 1239.5$ nm). b) Multipole decomposition of the scattering spectra for a particle in the metasurface with electric mirror effect. c) Visualization of the π -phase shift ($\lambda/2$, where $\lambda = 1550$ nm) in reflection between the metasurface supporting magnetic mirror effect (left panel) and the metasurface of electric mirror effect (middle panel). Their comparison with the perfect electric conductor (PEC) placed at the center of the nanoparticles.

electric mirror leads to destructive interference in the near field region. The near field of magnetic mirror due to the constructive interference has its maximum. The operation of these two mirrors differs in the phase control of the reflected beam, where the phase of the total electric field can be modulated around half of the wavelength $\lambda/2$.

Our precise numerical calculations show that the difference between the two marginal cases does not align exactly with 180° , but instead displays a slight deviation, reaching 2.5° . This discrepancy can be attributed to the influence of higher order modes in reflectivity and limitations in achieving complete elimination of the MD response. Furthermore, we conducted a comparative analysis of metasurfaces with a perfect electric conductor (PEC) positioned at the center of the nanoparticles. This comparison indicated that the effective center of reflection for these nanoparticles is displaced relative to their physical center, which results in a consequential shift of the far-field up to 20° .

4.2. Other Cases

A comprehensive overview of all potential configurations of a metasurface, with varying aspect ratios of nanodisks to achieve a high reflective state in the telecom wavelength range, is presented in Figure 5a. It's important to note that a general analysis of the spectral representation is essential for a proper understanding of all possible resonant states and their spectral positions relative

to each other. By adjusting the dimensional parameters of the nanoparticle array, each resonant state can be tuned to a desired wavelength. In our case, the radius of the disks and their periodicity ($R = 337$ nm, $P = 1239.5$ nm) are taken from the previous optimization analysis, where the magnetic mirror was specifically designed to exhibit high reflectivity at 1550 nm wavelength. All metasurfaces are configured in the non-diffractive regime, meaning the periodicity P is smaller than the wavelength λ . For this analysis, we still consider air surrounding the structures ($n_d = 1$). The refractive index of silicon n_{Si} in the further investigation is taken as the wavelength-dependent value at room temperature according to Ref. [28].

Total reflection, represented by the red area, is observed across various nanoparticle resonances and their combinations. By choosing the specific aspect ratio of nanodisks and looking at the spectra of multipole decomposition, we can define multipole moments responsible for the optical properties of the metasurface.

In Figure 5b, we illustrate an example of a metasurface with the arbitrarily chosen aspect ratio of $H/R = 1.6$. The purple dashed line represents the reflection profile, with its corresponding axis on the right side of the graph. The scattering cross-section of multipoles is presented by solid lines with its axis on the left side. The metasurface, characterized by dimensional parameters $R = 337$ nm, $H/R = 1.6$, and $P = 1239.5$ nm, exhibits full reflection at the wavelengths of 1511, 1636, and 1912 nm. The first point of total reflection at the lowest wavelength occurs due to the electric quadrupole moment (EQ), followed by the

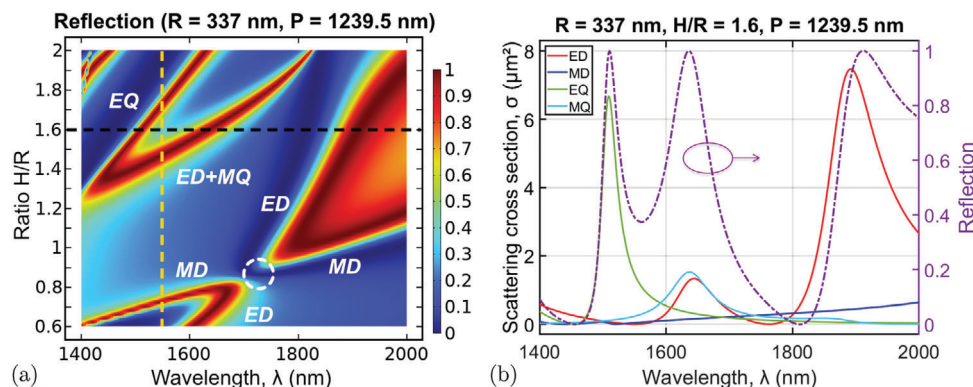


Figure 5. a) Reflection of metasurfaces with different aspect ratios for the telecom wavelength range. The 2D plot shows the relative spectral position of the dipole and quadrupole moments responsible for the high reflectivity of the metasurfaces. The dashed white circle indicates the area of high transmittance due to the overlap of the magnetic and electric dipole moments (Kerker effect). The dashed black line is a reflection spectrum presented in (b). The dashed orange line shows possible high-reflective metasurface configurations at 1550 nm wavelength. b) Multipole decomposition (left axis) and reflection (right axis) of the metasurface with $H/R = 1.6$ aspect ratio at telecom wavelengths.

second maximum resulting from the mutual contribution of electric dipole (ED) and magnetic quadrupole (MQ) moments. The last reflection maximum is attributed to ED with a small contribution of MD. By adjusting the disk's radius and periodicity while maintaining a constant aspect ratio, we can tailor each resonance to the desired wavelength.

Furthermore, Figure 5a shows that achieving total reflection at a wavelength of 1550 nm is not limited to the magnetic mirror regime corresponding to the aspect ratio $H/R = 0.725$. Total reflection is also attainable for $H/R = 1.48$ with the excitation of ED and MQ moments (dashed orange line), as well as for $H/R = 1.72$ with the dominant contribution of the EQ moment.

Within the aspect ratio range of 0.795 to 1, the overlap between ED and MD resonances precludes high reflectivity of the metasurfaces at any wavelength (the white dashed circle in Figure 5a). As mentioned above, the phenomenon, known as the Kerker effect, originates from the destructive interference of scattered light from ED and MD moments and can lead to the Huygens metasurface effect.^[13,25] The reflection bandgap between the marginal points of aspect ratios $H/R = 0.795$ (dashed) and $H/R = 1$ (solid) is shown in Figure 6. The mutual crossover of ED and MD resonances is clearly visible. The MD resonance undergoes a more rapid spectral shift compared to the ED resonance, as detailed in the earlier sections. It is interesting to notice that the maxima of reflection in Figure 6 align with the intersection points of ED and MD resonances, rather than with the maxima of their peaks. The values of the scattering cross section at these points from both resonances are equal, resulting in a comparable contribution to perfect reflection. Extracting information about ED and MD moments and substituting it in Equations (3) and (4), we observe that these metasurface configurations are close to the third partial case Equation (10).

The next analytical step, validating the concept for practical realization, involves the consideration of a metasurface within a medium.

5. Metasurface in Medium

For practical realization, let's consider a metasurface placed in the surrounding medium with the refractive index $n_d = 1.4$ (cor-

responding, for example, to polydimethylsiloxane^[29] and fused silica^[30] materials). To ensure operation in the non-diffractive regime, the periodicity of the nanodisks must be smaller than λ/n_d . For comparison, let's assume the same nanodisk radius of $R = 337$ nm, but with the scaling periodicity of $P/n_d = 885.4$ nm, where P is the periodicity in Figure 5a.

The reflection plot in Figure 7a for different nanodisk aspect ratios and wavelengths demonstrates the potential to attain total reflection, analogous to the high-reflective curves observed in Figure 5a. Note that the contributions of quadrupole moments in the scattering are weak for nanodisks of this radius.

The dashed black line corresponds to a reflection profile for the metasurface with an aspect ratio of $H/R = 0.725$, which was the parameter for the magnetic mirror effect in Section (3.2). The multipole decomposition of the metasurface (Figure 7b) illustrates the position of the MD resonance maximum with respect to the anapole state. Their relative shift (absence of overlap) indicates an uneven displacement of the resonant curves due to

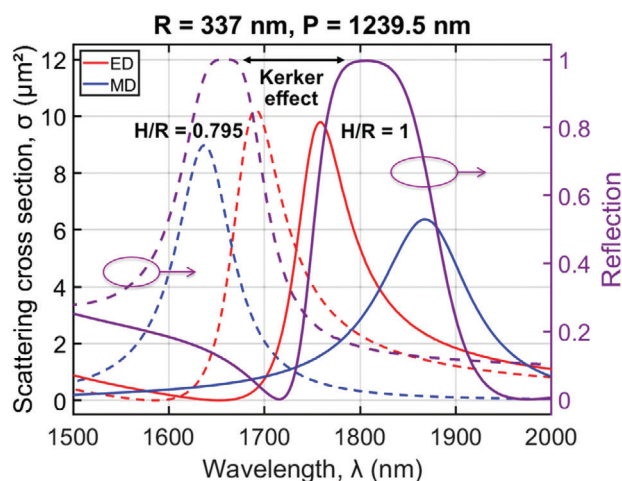


Figure 6. Scattering cross-section of electric and magnetic dipole moments (left axis) and reflection (right axis) for two metasurfaces ($H/R = 0.795$ & $H/R = 1$) as functions of wavelength.

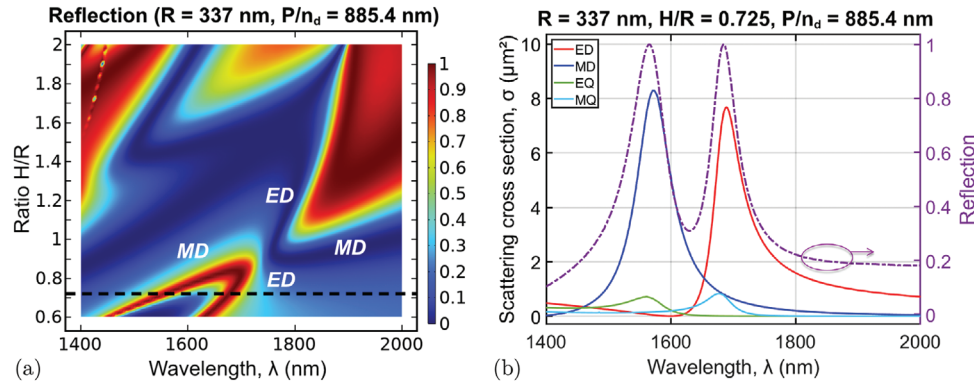


Figure 7. a) Reflection of metasurfaces embedded in a medium with refractive index $n_d = 1.4$ for different aspect ratios in the telecom wavelength range. The reflection profile at $H/R = 0.725$ is marked by the black dashed line. b) Multipole decomposition (left axis) and reflection (right axis) of the metasurface with $H/R = 0.725$ aspect ratio at telecom wavelengths. The wavelength λ is presented in free space.

the surrounding medium. The behavior originates from different field redistribution in MD and ED resonances. To obtain a magnetic mirror effect in the medium with $n_d = 1.4$, another optimization iteration is necessary, following the same steps described in the previous chapters.

The reflection profile is represented by a purple dashed line with the axis on the right side (Figure 7b). Although the observed dipole resonances don't resemble the specific cases of Equation (7) because of the contribution of the quadrupole moments, the total reflection is achievable for the general scenario.

Under fabrication considerations, the silicon nanodisk metasurface structure can be fabricated on top of a dielectric substrate, which can affect metasurface optical properties.^[31,32] By covering the nanostructure with an additional matching layer with a refractive index close to that of the substrate, the metasurface can be considered in a homogeneous environment. Therefore, in the experimental realization, the near-complete reflection can be achieved in accordance with the models presented in our paper.

The additional estimations show that reflectivity can remarkably depend on polarization of oblique incident light field. However, for the gravitational wave detection application, the linearly polarized light at zero angle of incidence is of primary interest, so our research is focused on this case.

6. Conclusion

In this paper, we have discussed nanodisk metasurfaces and explored their potential operational configurations to achieve high reflectivity. We presented a concept of how to step-by-step optimize the dimensional parameters of the metasurface and tune it to the specific resonant response at the wavelength of interest.

As an example of the design, a reflective metasurface operating at the magnetic dipole moment aligned to the anapole state (magnetic mirror effect) at a wavelength of 1550 nm has been proposed. This wavelength was chosen based on the potential implementation of the technology in the research field of gravitational wave detection. Successive tuning of the aspect ratio and radius of an individual nanodisk and then the periodicity of the array demonstrated the precise optimization of the ideal reflection effect at the required wavelength. At the same time, informa-

tion on the multipole contribution to the reflectivity was formally obtained from single-particle cross-section simulation with multipole decomposition. For the same wavelength of 1550 nm we were able to numerically modify the radius of the disks so that they would be resonant at electric dipole moment.

We showed various metasurface designs capable of achieving high reflection with varying phases of the reflected light field in the range of π to $-\pi$, accomplished through a combination of dipoles. Additionally, we demonstrated the potential of high reflection due to the influence of quadrupole moments. Notably, even in the scenario of the metasurface embedded in the surrounding medium with the refractive index of $n_d = 1.4$, full reflection was successfully achieved in the simulations. The considered theory and simulations are in full agreement with each other. The developed approach can be applied to the tuning of the perfect mirror effect at different wavelengths in the spectral region where the absorption in silicon is negligibly small. The results pave the way for the potential implementation of this research in future fabrication processes.

Appendix A: Multipole Decomposition of Scattering Cross-Sections

The total scattering cross-section is a sum of contributions from different multipole moments^[33–35]:

$$\sigma_{\text{sca}}^{\text{total}} = \sigma_{\text{sca}}^p + \sigma_{\text{sca}}^m + \sigma_{\text{sca}}^{Q^e} + \sigma_{\text{sca}}^{Q^m} + \dots \quad (\text{A1})$$

where p - the electric dipole moment, m - the magnetic dipole moment, Q^e - the electric quadrupole moment, Q^m - the magnetic quadrupole moment.

This expression can also be written as follows:

$$\sigma_{\text{sca}}^{\text{total}} = \frac{k_0^4}{6\pi\epsilon_0^2|E_{\text{inc}}|^2} \sum_{\alpha} |p_{\alpha}|^2 + \frac{k_0^4\epsilon_d\mu_0}{6\pi\epsilon_0|E_{\text{inc}}|^2} \sum_{\alpha} |m_{\alpha}|^2 + \frac{k_0^6\epsilon_d}{720\pi\epsilon_0^2|E_{\text{inc}}|^2} \sum_{\alpha\beta} |Q_{\alpha\beta}^e|^2 + \frac{k_0^6\epsilon_d^2\mu_0}{80\pi\epsilon_0|E_{\text{inc}}|^2} \sum_{\alpha\beta} |Q_{\alpha\beta}^m|^2 + \dots \quad (\text{A2})$$

where E_{inc} is the electric field of the incident wave, k_0 is the wavenumber in vacuum, ϵ_0 and μ_0 are the electric permittivity

and the magnetic permeability of free space, respectively, ϵ_d is the relative dielectric permittivity of the surrounding medium.

The exact multipole moments can be derived using the induced electric current density $\mathbf{J}^\omega$, distributed within the nanoparticle volume V . The indices α, β iterate over the values x, y, z .

$$p_\alpha = \frac{i}{\omega} \left\{ \int_V d^3\mathbf{r} \mathbf{J}_{\alpha 0}^\omega(k_d r) + \frac{k_d^2}{2} \int_V d^3\mathbf{r} [3(\mathbf{r} \cdot \mathbf{J}^\omega) r_\alpha - r^2 J_\alpha^\omega] \frac{j_2(k_d r)}{(k_d r)^2} \right\} \quad (\text{A3})$$

$$m_\alpha = \frac{3}{2} \int_V d^3\mathbf{r} [\mathbf{r} \times \mathbf{J}^\omega]_\alpha \frac{j_1(k_d r)}{k_d r} \quad (\text{A4})$$

$$Q_{\alpha\beta}^e = \frac{3i}{\omega} \left\{ \int_V d^3\mathbf{r} [3(r_\alpha J_\beta^\omega + r_\beta J_\alpha^\omega) - 2(\mathbf{r} \cdot \mathbf{J}^\omega) \delta_{\alpha\beta}] \frac{j_1(k_d r)}{k_d r} + 2k_d^2 \int_V d^3\mathbf{r} [5r_\alpha r_\beta (\mathbf{r} \cdot \mathbf{J}^\omega) - (r_\alpha J_\beta^\omega + r_\beta J_\alpha^\omega) r^2 - r^2 (\mathbf{r} \cdot \mathbf{J}^\omega) \delta_{\alpha\beta}] \frac{j_3(k_d r)}{(k_d r)^3} \right\} \quad (\text{A5})$$

$$Q_{\alpha\beta}^m = 15 \int_V d^3\mathbf{r} (r_\alpha [\mathbf{r} \times \mathbf{J}^\omega]_\beta + r_\beta [\mathbf{r} \times \mathbf{J}^\omega]_\alpha) \frac{j_2(k_d r)}{(k_d r)^2} \quad (\text{A6})$$

k_d is the wavenumber in the surrounding medium, $j_n(k_d r)$ is the n -th order spherical Bessel function.

The simulation software Comsol Multiphysics, used in the paper for the metasurface optimization process, calculates the polarization density $\mathbf{P}$, which is related to the electric current density $\mathbf{J}^\omega$ by the equation:

$$\mathbf{P} = -\frac{\mathbf{J}^\omega}{i\omega} \quad (\text{A7})$$

The information about the total scattering cross-section is applicable to the study of resonances of a single nanoparticle. In the context of metasurface investigation, our focus shifts to the relative contributions of individual multipoles. By extracting information about the polarization density $\mathbf{P}'$ of the nanoparticle within the array and incorporating this data into the equations for multipole decomposition, we can estimate the relative excitation of dipoles and quadrupoles across the spectral range.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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anapole, coatings, dielectric metasurface, metamaterials, mirror effect, multipole response, optical nanostructures

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